

SOHAR MAJIS, Oman

WMO: 412460

Lat: 24.4674N

Lon: 56.6445E

Elev: 7

StdP: 14.69

Time Zone: 4.00 (E04)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	56.1	58.5	34.1	28.8	80.3	38.5	34.4	77.3	20.2	72.3	17.6	73.2	5.3	240	0.366

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	11.2	104.7	74.3	100.7	79.0	98.6	81.4	89.8	95.5	88.5	94.6	87.5	93.8	9.5	60

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
88.5	206.9	94.2	87.2	198.0	93.3	86.0	190.8	92.7	55.9	95.7	54.2	94.6	52.9	94.0	96.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
16.9	14.6	13.3	DB	52.4	112.9	3.6	3.3	49.8	115.3	47.7	117.2	45.7	119.1	43.1	121.5
			WB	44.8	90.5	3.0	2.6	42.6	92.3	40.8	93.8	39.1	95.2	36.9	97.1

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	81.5	68.1	69.7	74.2	81.9	89.3	92.2	92.0	90.2	87.7	83.9	76.7	71.0	(6)
	DBStd	9.39	3.18	3.68	4.04	4.80	4.47	3.53	2.66	2.91	2.60	3.38	3.38	3.14	(7)
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
	HDD65	14	8	5	0	0	0	0	0	0	0	0	0	1	(9)
	CDD50	11487	562	551	750	957	1218	1267	1303	1246	1130	1050	802	651	(10)
	CDD65	6028	105	136	286	507	753	817	838	781	680	585	353	187	(11)
	CDH74	80649	349	581	2109	6102	11616	13223	13321	11824	9822	7550	3170	982	(12)
	CDH80	46007	17	45	422	2757	7280	8907	8857	7365	5585	3790	906	76	(13)
(14) Wind	WSAvg	5.8	5.8	5.9	5.9	5.9	5.8	5.9	6.2	6.2	5.6	5.1	5.4	5.5	(14)
(15) Precipitation	PrecAvg	3.7	0.5	0.8	0.8	0.3	0.0	0.0	0.1	0.1	0.0	0.2	0.3	0.4	(15)
	PrecMax	12.3	2.9	3.6	3.1	1.3	0.1	0.2	2.2	0.2	0.2	4.4	2.2	6.7	(16)
	PrecMin	1.0	0.0	0.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	(17)
	PrecStd	2.6	0.7	0.8	1.0	0.4	0.0	0.0	0.4	0.0	0.0	0.8	0.4	1.3	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	82.1	84.6	92.1	102.3	109.1	111.1	105.9	104.3	100.3	97.2	90.0	85.1	(19)
		MCWB	64.8	65.9	67.3	68.0	69.8	70.3	78.8	79.4	77.5	74.6	72.7	66.1	(20)
	2%	DB	78.8	80.8	86.6	96.5	103.6	104.9	100.6	99.1	96.3	94.7	87.8	81.8	(21)
		MCWB	66.9	68.4	68.9	70.4	70.1	75.4	82.8	83.5	81.0	77.2	74.3	68.9	(22)
	5%	DB	77.2	78.9	83.9	93.0	99.1	100.4	98.3	96.8	94.6	93.0	86.3	80.2	(23)
		MCWB	66.8	68.4	70.4	73.9	76.6	81.7	84.6	84.5	82.5	78.3	74.2	69.0	(24)
	10%	DB	75.9	77.3	81.9	90.3	96.7	98.2	96.7	95.0	93.1	91.3	84.9	78.9	(25)
		MCWB	66.1	68.1	70.3	74.4	78.6	82.8	85.2	85.0	83.0	78.6	73.5	68.5	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	72.4	74.3	77.5	83.4	89.9	93.1	91.4	90.3	88.0	85.8	79.4	75.1	(27)
		MCDB	77.5	79.0	83.1	90.1	96.2	97.8	95.5	96.7	93.8	91.8	86.2	80.0	(28)
	2%	WB	70.4	72.8	75.5	81.0	86.5	89.9	89.5	88.9	86.3	83.6	77.6	72.7	(29)
		MCDB	76.2	77.9	81.2	88.6	93.9	95.3	95.1	95.1	92.2	90.4	85.2	79.0	(30)
	5%	WB	69.1	71.6	74.2	79.5	85.0	88.0	88.4	87.7	85.4	82.2	76.2	71.2	(31)
		MCDB	75.3	76.8	80.2	87.1	92.9	94.6	94.3	93.4	91.4	89.5	84.3	78.0	(32)
	10%	WB	67.7	70.2	73.0	78.0	83.6	86.8	87.5	86.6	84.5	80.8	74.8	69.6	(33)
		MCDB	74.4	75.7	79.2	85.8	92.2	94.0	93.7	92.4	90.7	88.7	83.1	76.8	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	14.8	14.7	14.1	14.3	13.9	11.2	8.2	8.7	10.9	14.7	15.1	14.9	(35)
		MCDBR	15.5	15.6	16.6	17.9	17.3	15.9	11.8	12.4	13.2	15.6	16.0	16.3	(36)
	5% WB	MCWBR	9.7	10.7	12.9	13.9	15.4	15.1	10.8	9.5	10.5	13.7	10.6	10.0	(37)
		MCWBR	14.4	12.7	12.6	14.1	13.2	11.6	9.1	9.3	10.8	13.1	15.1	14.5	(38)
(40) Clear-Sky Solar Irradiance	taub		0.437	0.492	0.551	0.588	0.594	0.659	0.739	0.694	0.595	0.502	0.452	0.435	(40)
		taud	2.169	2.001	1.847	1.792	1.777	1.678	1.587	1.661	1.815	2.004	2.151	2.171	(41)
	Ebn at Noon		256	250	242	236	233	217	200	209	227	243	249	251	(42)
		Edh at Noon	43	54	66	71	71	78	86	80	67	53	43	41	(43)
(44) All-Sky Solar Radiation	RadAvg		1372	1610	1890	2138	2374	2382	2212	2122	2000	1773	1458	1301	(44)
	RadStd		83	103	118	146	68	82	81	65	40	43	34	69	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (16 neighbors)	+1.22	+2.92	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		+0.72	+1.10	N/A	N/A	N/A	N/A	N/A	N/A	+230	+227

Nomenclature: See separate page